Filtering the 2023-2024 dataset for Aphis fabae

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Summary

Progressive filtering of the dataset. After each filtering steps different metrics are measured to asses the dataset refinement. These key metrics include broad sence heritability and normality of the data.

Setup

Packages

This analysis depends on the following packages:

```
suppressPackageStartupMessages(library("tidyverse"))
suppressPackageStartupMessages(library("lme4"))
suppressPackageStartupMessages(library("Hmisc"))
suppressPackageStartupMessages(library("corrplot"))
suppressPackageStartupMessages(library("emmeans"))
suppressPackageStartupMessages(library("patchwork"))
```

Helper functions

Two helper functions are created:
- `calculate_broad_heritability()` giving the broad sense heritability (H^2) and the number of genotypes of a given dataset.
- `create_correlation_plot()` to calculate the correlations between experimental sessions.

```
source("../../broad_heritability_function.R")
source("../../create_correlation_plot.R")
```

```
Loading required package: glue
```

Broad-sense heritability calculations

Raw dataset

```
data250_afabae_00 <- read.csv("../../../../../../01.data/2023_screen_specifics_251genotypes/0
  mutate(ivip = as.factor(ivip)) %>%
  mutate(session = as.factor(session))

res <- calculate_broad_heritability(data = data250_afabae_00)
```

we can't use our dataset like this, we need to clean it first.

> Tip
>
> Conclusions:
> - Heritability is equal to 0.102.
> - There are 220 genotypes in total.
> - There are 2652 measurements in total.

Missing data

First step we remove all line with a NA for larvae count.

```
data250_afabae_01 <- data250_afabae_00 %>% filter(!is.na(larvae_total_number))
data250_afabae_01$ivip <- droplevels(data250_afabae_01$ivip)

res2 <- calculate_broad_heritability(data = data250_afabae_01)
```

> **Tip**
>
> Conclusions:
> - Heritability is equal to 0.102.
> - There are 218 genotypes in total.
> - There are 2358 measurements in total.

```
create_correlation_plot(data250_afabae_01,
                        group_cols = c("genotype", "session"),
                        value_col = "larvae_total_number")
```

Remove observations with no adult alive

We chose to remove all data from plants on which no adult aphid was found. This decision was made to get rid of larvae installation bias and only assess the reproduction ability of the aphid.

When the adult aphid dies, the number of larvae will drop but only because the adult stayed spent less time on the plant and not because the plant was not suitable for the reproduction.

```
data250_afabae_02<- data250_afabae_01 %>% filter(adult_alive != "0") #remove if no adult
data250_afabae_02$ivip <- droplevels(data250_afabae_02$ivip)

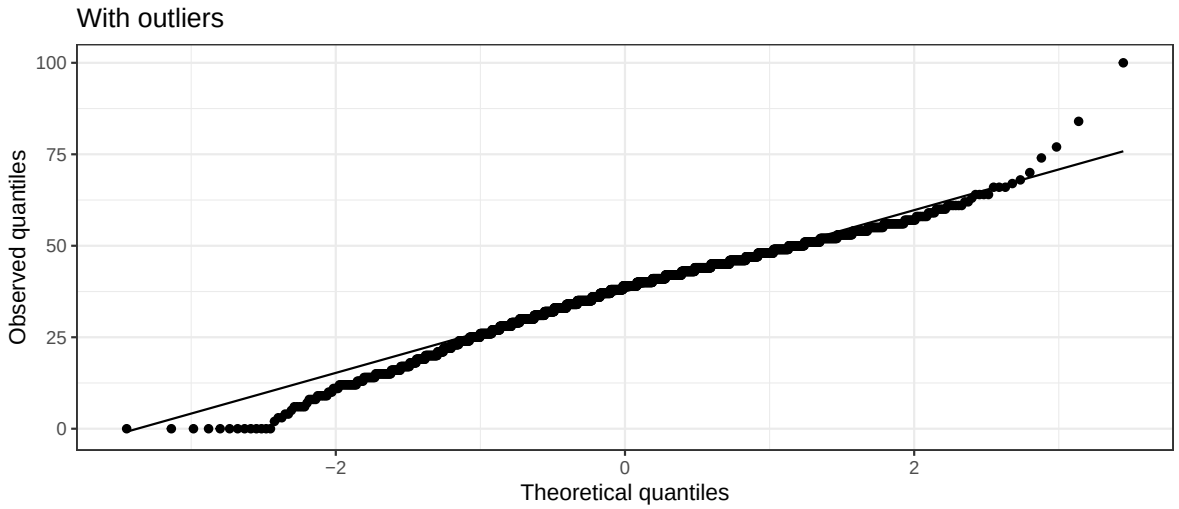
res3 <- calculate_broad_heritability(data = data250_afabae_02)
```

> **Note**
>
> Conclusions:
> - Heritability is equal to 0.294.
> - There are 218 genotypes in total.
> - There are 1759 measurements in total.

Heritability strongly increases. 602 measurements are lost. The number of genotype stays the same.

```
create_correlation_plot(data250_afabae_02,
                        group_cols = c("genotype", "session"),
                        value_col = "larvae_total_number")
```

```
# With outliers
p_with <- data250_afabae_02 %>%
  ggplot(mapping = aes(sample=larvae_total_number)) +
  stat_qq() +
  stat_qq_line() +
  labs(y = "Observed quantiles", x = "Theoretical quantiles") +
  ggtitle("With outliers") +
  theme_bw()
p_with
```



We observ few outlier data on both side of the distribution. To remove these data we use the Interquartile range method (IQR). This will help inproving normality of the data.

Outliers removal

```
x<-1.5
IQR_value <- IQR(data250_afabae_02$larvae_total_number)
lower_bound <- quantile(data250_afabae_02$larvae_total_number, 0.25) - x * IQR_value
```

```r
upper_bound <- quantile(data250_afabae_02$larvae_total_number, 0.75) + x * IQR_value

# Filter out outliers
data250_afabae_03 <- data250_afabae_02[data250_afabae_02$larvae_total_number >= lower_bound &

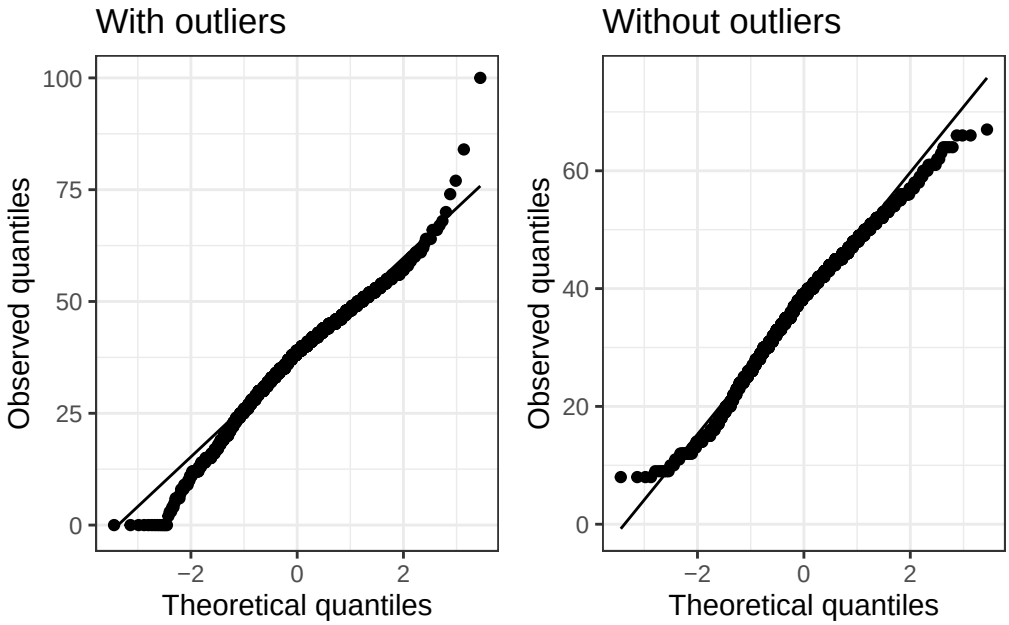
# Identify the rows that were removed (outliers)
removed_outliers <- data250_afabae_02[data250_afabae_02$larvae_total_number < lower_bound | d

res4 <- calculate_broad_heritability(data = data250_afabae_03)


#| fig-width: 8

# Without outliers
p_without <- data250_afabae_03 %>%
  ggplot(mapping = aes(sample=larvae_total_number)) +
  stat_qq() +
  stat_qq_line() +
  labs(y = "Observed quantiles", x = "Theoretical quantiles") +
  ggtitle("Without outliers") +
  theme_bw()

p_with + p_without
```



! Important

Conclusions:
- Heritability is equal to 0.343.
- There are 218 genotypes in total.
- There are 1728 measurements in total.

Final filtering: keep genotypes with at least 4 repetitions

To make sure the data we use for the GWAS are reliable we set a threshold for the minimal number of repetition to keep a genotype.

```
data250_afabae_04_4 <- data250_afabae_03 %>%
  group_by(ivip) %>%           # Group by genotype
  filter(n() >= 4) %>%             # Keep only groups with 6 or more repetitions
  ungroup()
data250_afabae_04_4$ivip <- droplevels(data250_afabae_04_4$ivip)

res5_4 <- calculate_broad_heritability(data = data250_afabae_04_4)
```

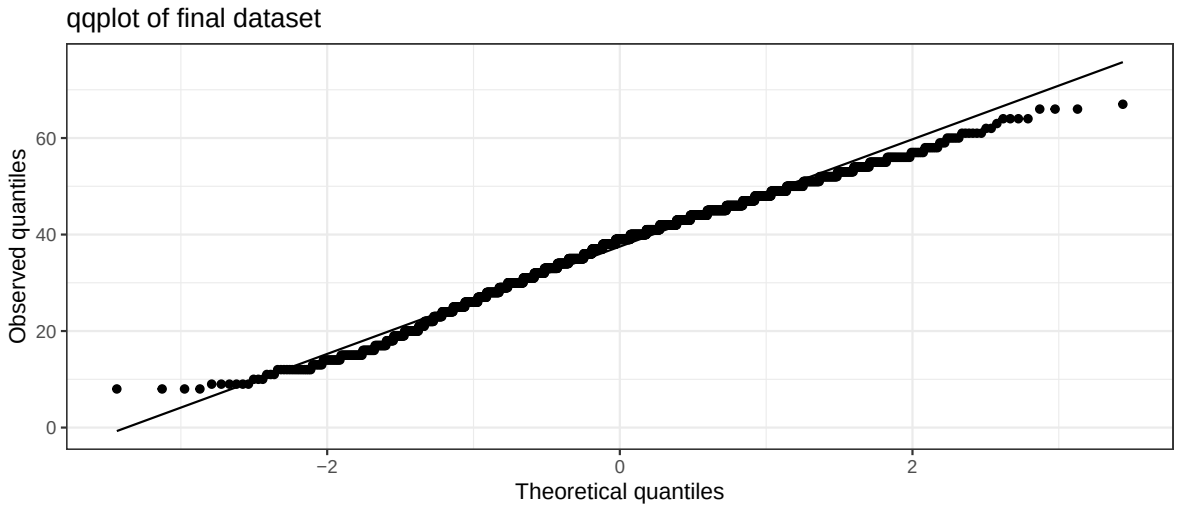
```
data250_afabae_04_5 <- data250_afabae_03 %>%
  group_by(ivip) %>%           # Group by genotype
  filter(n() >= 5) %>%             # Keep only groups with 6 or more repetitions
  ungroup()
data250_afabae_04_5$ivip <- droplevels(data250_afabae_04_5$ivip)

res5_5 <- calculate_broad_heritability(data = data250_afabae_04_5)
```

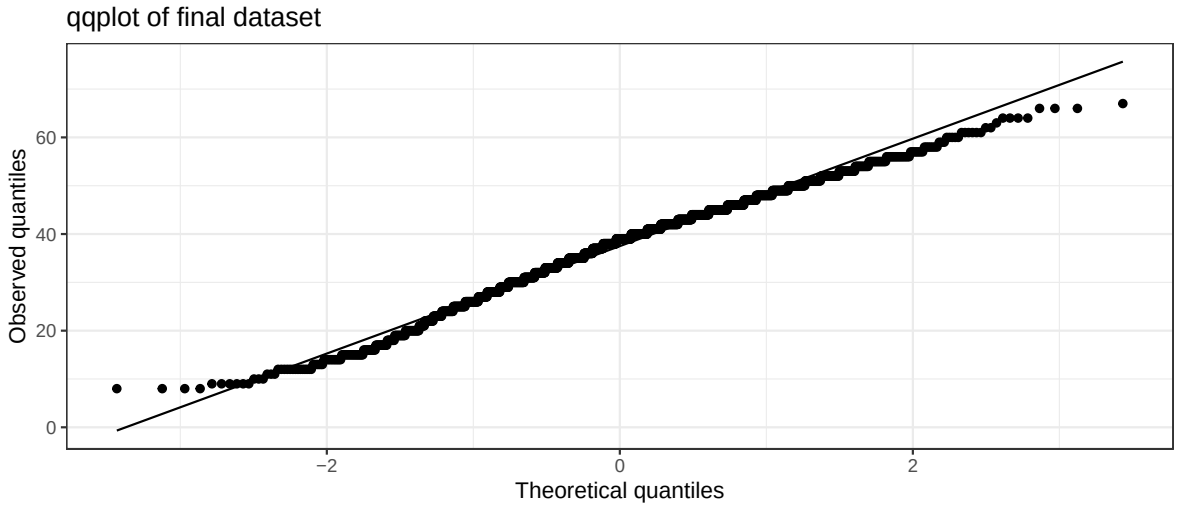
```
data250_afabae_04_6 <- data250_afabae_03 %>%
  group_by(ivip) %>%           # Group by genotype
  filter(n() >= 6) %>%             # Keep only groups with 6 or more repetitions
  ungroup()
data250_afabae_04_6$ivip <- droplevels(data250_afabae_04_6$ivip)

res5_6 <- calculate_broad_heritability(data = data250_afabae_04_6)
```

```
p <- data250_afabae_04_4 %>%
  ggplot(mapping = aes(sample=larvae_total_number)) +
  stat_qq() +
  stat_qq_line() +
  labs(y = "Observed quantiles", x = "Theoretical quantiles") +
  ggtitle("qqplot of final dataset") +
  theme_bw()
p
```

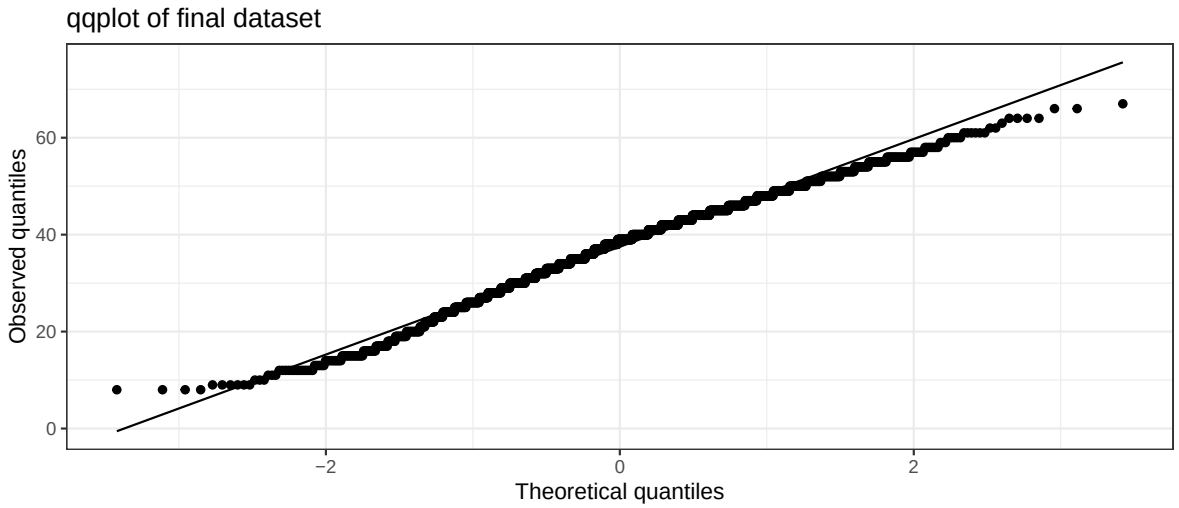



```
p <- data250_afabae_04_5 %>%
  ggplot(mapping = aes(sample=larvae_total_number)) +
  stat_qq() +
  stat_qq_line() +
  labs(y = "Observed quantiles", x = "Theoretical quantiles") +
  ggtitle("qqplot of final dataset") +
  theme_bw()
p
```




```
p <- data250_afabae_04_6 %>%
  ggplot(mapping = aes(sample=larvae_total_number)) +
```

```
  stat_qq() +
  stat_qq_line() +
  labs(y = "Observed quantiles", x = "Theoretical quantiles") +
  ggtitle("qqplot of final dataset") +
  theme_bw()
p
```

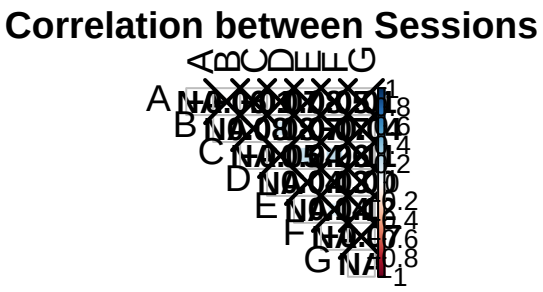


! Important

Conclusions:
- Heritability is equal to 0.341.
- There are 208 genotypes in total.
- There are 1705 measurements in total.

Analysis of the correlation between each session

```
cor<-create_correlation_plot(data250_afabae_04_4,
                             group_cols = c("genotype", "session"),
                             value_col = "larvae_total_number")
```




```
cortable<-cor$corrPos
mean(cortable$corr)
```

```
[1] 0.06161094
```

```
sd(cortable$corr)
```

```
[1] 0.09538721
```

```
mean(cortable$corr[cortable$p.value < 0.1], na.rm = TRUE)
```

```
[1] 0.2038454
```

```
sd(cortable$corr[cortable$p.value < 0.1], na.rm = TRUE)
```

```
[1] 0.0523963
```

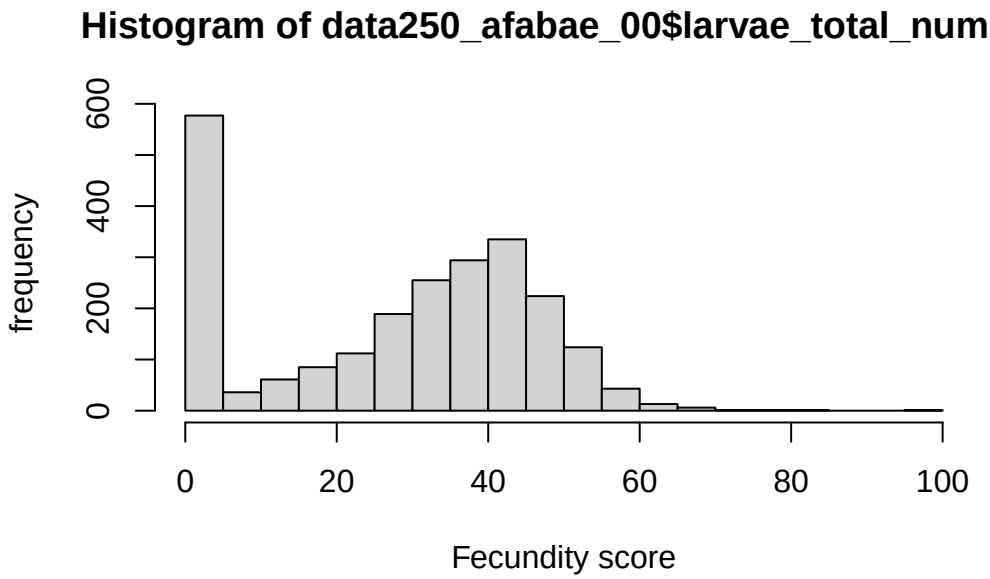
the correlation is low betwwen replicate

heritability:

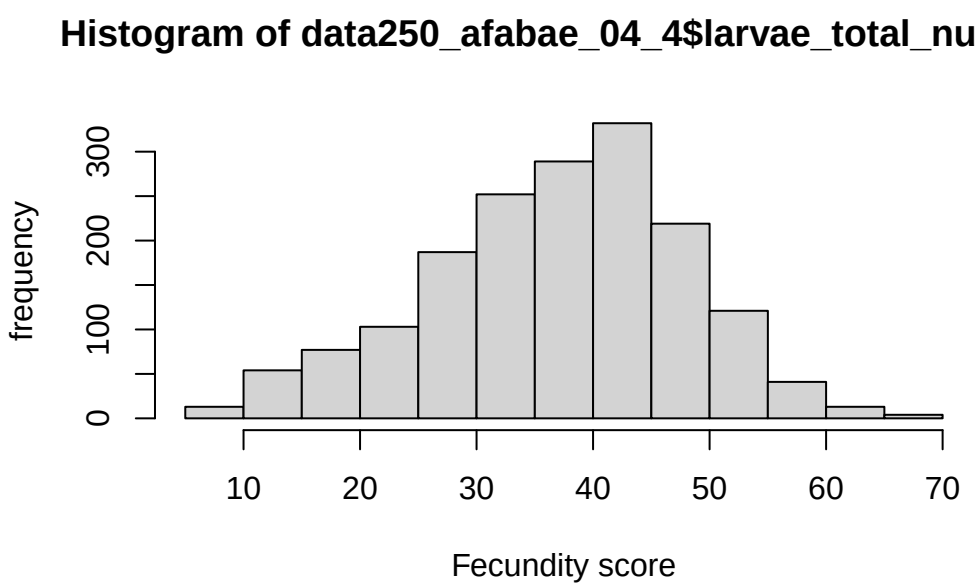
```r
mod_h2 <- lmer(larvae_total_number ~ (1 | ivip) + (1 | session), data = data250_afabae_04_4)
  estimates <- as.data.frame(VarCorr(mod_h2))
  genetic_variance <- estimates$vcov[1]
  residual_variance <- estimates$vcov[3]
  nbE <- nlevels(data250_afabae_04_4$session)
  H2 <- genetic_variance / (genetic_variance + (residual_variance / nbE))
```

comparison between raw and final distribution

```r
hist(data250_afabae_00$larvae_total_number, breaks = 20, xlab = "Fecundity score", ylab = "fr
```




```r
hist(data250_afabae_04_4$larvae_total_number, breaks = 20, xlab = "Fecundity score", ylab = "
```



Write final results to .csv

```
write.csv(x = data250_afabae_04_4,
          file = "../../../01.dataset_filtering/01_A.fabae/final filters/aphis_fabae_data_fi
          quote = F, row.names = F)
```

```
write.csv(x = data250_afabae_04_5,
          file = "../../../01.dataset_filtering/01_A.fabae/final filters/aphis_fabae_data_fi
          quote = F, row.names = F)
```

```
write.csv(x = data250_afabae_04_6,
          file = "../../../01.dataset_filtering/01_A.fabae/final filters/aphis_fabae_data_fi
          quote = F, row.names = F)
```